

FOR SECTION I, BUBBLE ALL ANSWERS ON SCANTRON.
FOR SECTION II, WRITE ALL ANSWERS ON THIS EXAM.

20 pts total

SECTION I: MULTIPLE CHOICE

SELECT SOLELY ONE ANSWER PER QUESTION ON THE PROVIDED SCANTRON.

15 QUESTIONS – 15 POINTS – 15 MINUTES

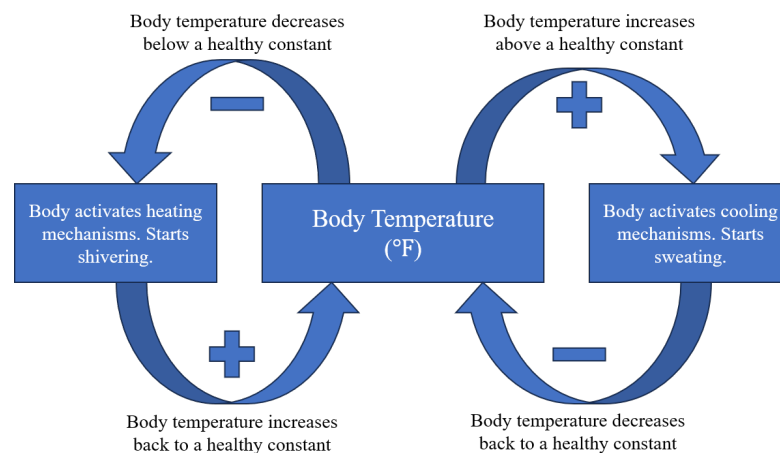
pacing recommendation: 1 minute / question

Ecology 1:

1. Which of the following are TRUE in relation to the diagram below?

- I: The diagram depicts a positive feedback mechanism.
- II: The diagram depicts a negative feedback mechanism.
- III: The diagram depicts homeostasis incorrectly.
- IV: The diagram does not represent homeostasis in any way.

- A. I and III
- B. II and III
- C. I and IV
- D. II only



2. Which of the following statements is TRUE in regard to homeostasis?

- A. Homeostasis involves maintaining relatively stable internal conditions that fluctuate within a bounded range, often differing from the external environment.
- B. Homeostasis ensures organisms are in complete equilibrium with their external environment, maintaining identical internal and external conditions.
- C. Homeostasis refers to a state where internal conditions are unstable but in equilibrium with the external environment's fluctuations.
- D. In general, homeostasis is maintained through positive feedback mechanisms.

3. What is the ultimate source of energy for all life on Earth?

- A. Decomposers
- B. Heterotrophs
- C. Autotrophs
- D. None of the above

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4. As energy is transferred throughout the food chain, which of the following occurs?
- A. Energy is converted into organic matter.
 - B. Energy is released as heat.
 - C. Energy is destroyed.
 - D. Organic matter is converted into energy.
5. Which of the following are utilized by producers to obtain chemical potential energy?
- I: Photosynthesis
 - II: Cellular Respiration
 - III: Chemosynthesis
- A. I and III – since producers can obtain chemical potential energy under exposure to sunlight and without in some cases.
 - B. II only – since cellular respiration breaks down glucose and other biomolecules in order to obtain usable energy in the form of ATP.
 - C. I only – since only autotrophs can perform chemosynthesis
 - D. None of the above
6. Which of the following is likely to happen if producers were fully removed from a particular ecosystem?
- I: Heterotroph populations would likely increase.
 - II: Heterotroph populations would likely decline.
 - III: Apex predator populations would likely decline.
 - IV: Secondary consumer populations would likely increase since secondary consumers are not directly dependent upon producers for energy.
- A. I and IV
 - B. II and IV
 - C. II and III
 - D. None of the above
7. Which of the following is likely to happen if secondary consumers were fully removed from a particular ecosystem?
- I: Autotroph populations would immediately seriously drop.
 - II: Primary consumer populations would immediately seriously drop.
 - III: Decomposer populations would immediately seriously drop.
 - IV: Tertiary consumer populations would soon seriously drop.
- A. I and IV
 - B. III and IV
 - C. I, II, III, and IV
 - D. None of the above

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8. Which of the following is an abiotic factor?

- A. Humidity
- B. Fallen Tree Leaves
- C. Gopher Snakes
- D. Both A and B

Ecology 2:

9. Fill In The Blank: Expansive slight differences in organisms of the same species in a particular ecosystem indicates high _____.

- A. Genetic Diversity
- B. Species Diversity
- C. Ecosystem Diversity
- D. Both A and B

10. Which of the following are provisioning ecosystem services?

- I: Cycling Matter
- II: Food
- III: Shelter
- IV: Flood Control

- A. I, II, and III
- B. IV only
- C. II and III
- D. I only

11. Which of the following is an example of a keystone species?

- A. **Gray Wolf:** Controls herbivore populations, preventing overgrazing and thus protecting plant species diversity in forests.
- B. **White-Footed Mouse:** Forages on seeds and small insects, contributing to nutrient cycling in forest ecosystems.
- C. **Common Starling:** Feeds on fruits and insects, participating in seed dispersal and local food webs.
- D. **None of the above**

12. Which of the following best describes conservation science?

- A. The study of maximizing biodiversity by increasing the total number of species in an ecosystem regardless of their impact.
- B. The practice of understanding key species' roles and maintaining ecosystem services critical to humans and ecosystems.
- C. The process of eliminating invasive species to ensure that native species continue to dominate the ecosystem.
- D. The focus on preserving only top predators to maintain ecological balance and prevent ecosystem collapse.

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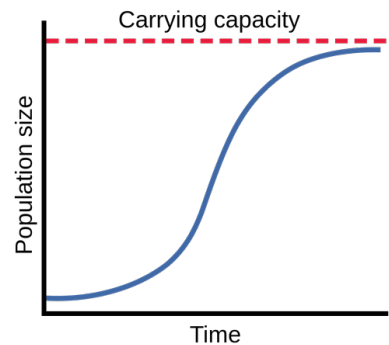
20 pts total

13. Which of the following are actions that could be taken to directly alleviate habitat loss?

- A. Building wildlife corridors
- B. Building riparian corridors
- C. Both A and B
- D. None of the above

14. Which of the following is TRUE in regard to the given graph?
(choose the best answer)

- A. This graph depicts population levels over time in an unrealistic manner.
- B. This graph depicts population levels over time in a realistic manner.
- C. Population growth could have subsided because of a high emigration rate and a very high birth rate.
- D. Population growth could have subsided because of a very low death rate and a high emigration rate.



15. The fact that Earth Overshoot Day is moving closer to the beginning of the year over time means that:

- A. We are borrowing resources from the past by decreasing resource availability to other organisms.
- B. We are increasing human carrying capacity.
- C. We are decreasing biodiversity in the present.
- D. None of the above

Section I ends.

SECTION II: FREE RESPONSE

ANSWER THE FOLLOWING FRQs TO THE BEST OF YOUR ABILITY. WRITE ON THIS EXAM.

1 QUESTION – 5 POINTS – 5 MINUTES

1. Fill In The Blanks: (each blank is worth 1 pt)

CHIPPO stands for Climate Change, Habitat Loss, _____, Pollution, Population, and _____.

A beautiful view of a rainforest is an example of a _____. (be specific!)

Most producers are able to perform _____ to obtain energy from the sun and also perform _____ to convert energy stored in biomolecules like glucose into usable energy in the form of ATP.

Section II ends.

- END OF EXAM -

Section II ends.